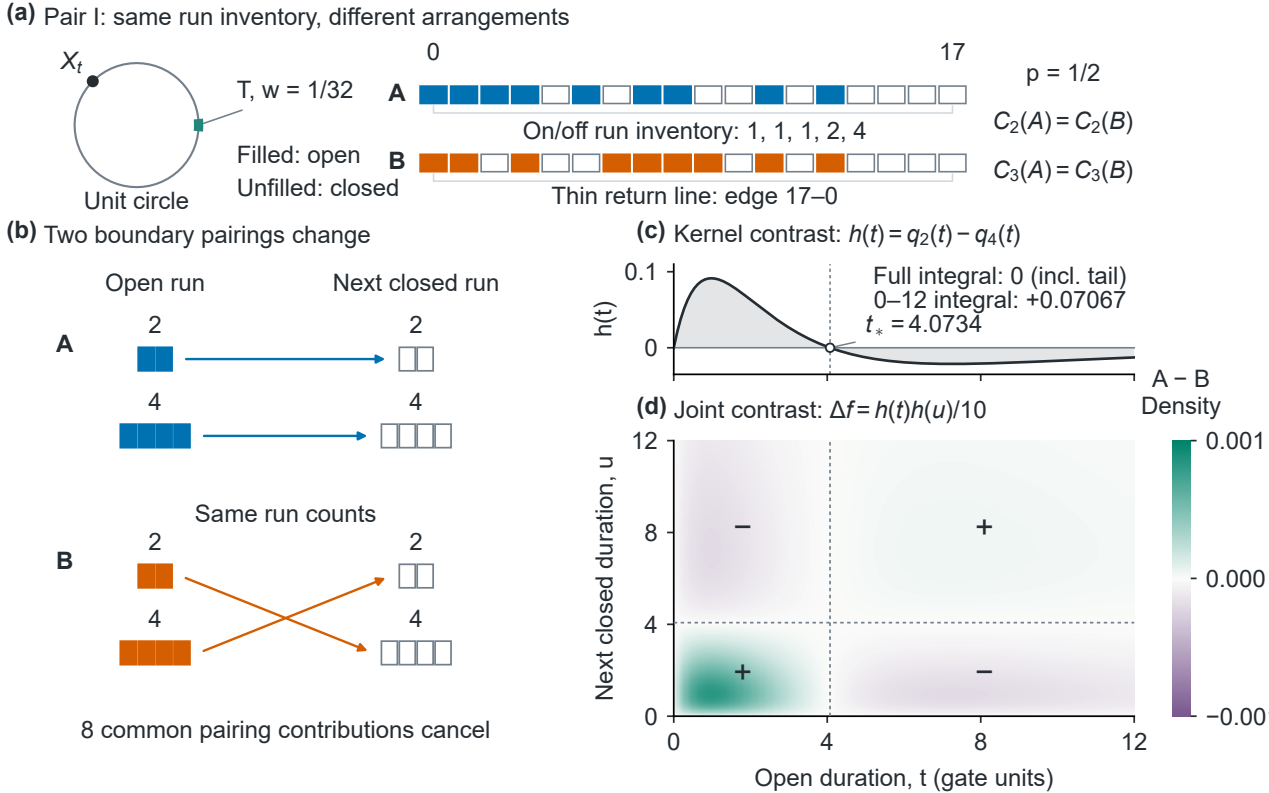

(a) Pair I: same run inventory, different arrangements
X_t
T, w = 1/32
Unit circle
Filled: open
Unfilled: closed
0
17
A
B
On/off run inventory: 1, 1, 1, 2, 4
Thin return line: edge 17–0
p = 1/2
$C_2(A) = C_2(B)$
$C_3(A) = C_3(B)$
(b) Two boundary pairings change
Open run
Next closed run
A
2
4
2
4
Same run counts
B
2
4
2
4
8 common pairing contributions cancel
(c) Kernel contrast: $h(t) = q_2(t) - q_4(t)$
0.1
0
h(t)
Full integral: 0 (incl. tail)
0–12 integral: +0.07067
$t_* = 4.0734$
A − B Density
(d) Joint contrast: $\Delta f = h(t)h(u)/10$
12
8
4
0
Next closed duration, u
−
+
+
−
0
4
8
12
Open duration, t (gate units)
0.001
0.000
−0.001